

Network Layer: Logical Addressing

IPv4 ADDRESSES

*An **IPv4 address** is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.*

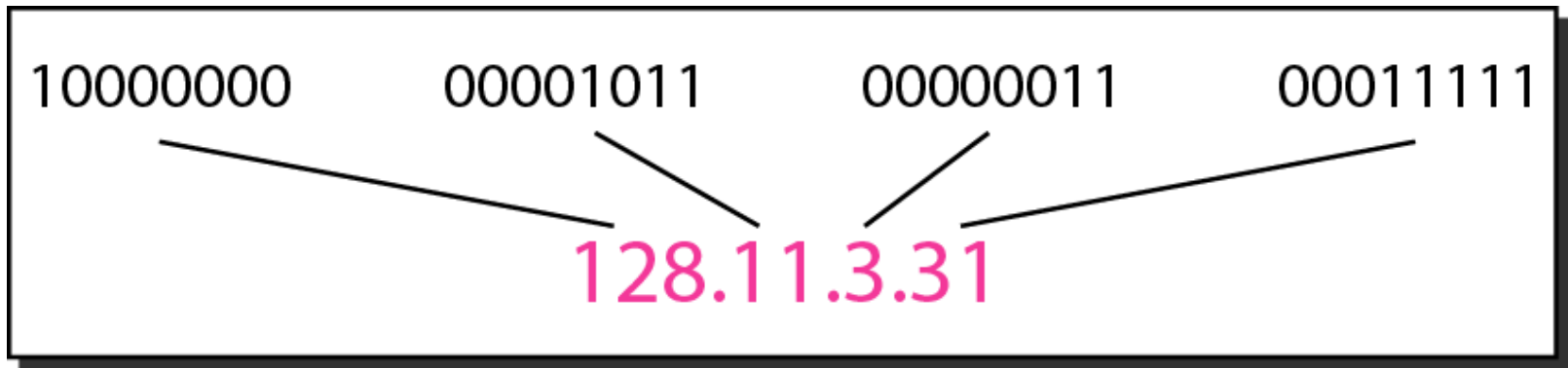
Note

An IPv4 address is 32 bits long.

The IPv4 addresses are unique and universal (all nodes connecting Internet must have IP addresses).

The address space of IPv4 is 2^{32} or 4,294,967,296.

Dotted-decimal notation and binary notation for an IPv4 address



Example

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Example

Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Example

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent.

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

Example

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Example

Solution

- a. There must be no leading zero (045).*
- b. There can be no more than four numbers.*
- c. Each number needs to be less than or equal to 255.*
- d. A mixture of binary notation and dotted-decimal notation is not allowed.*

Note

**In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.**

Finding the classes in binary and dotted-decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

Class D: multicast
Class E: reserved

Example

Find the class of each address.

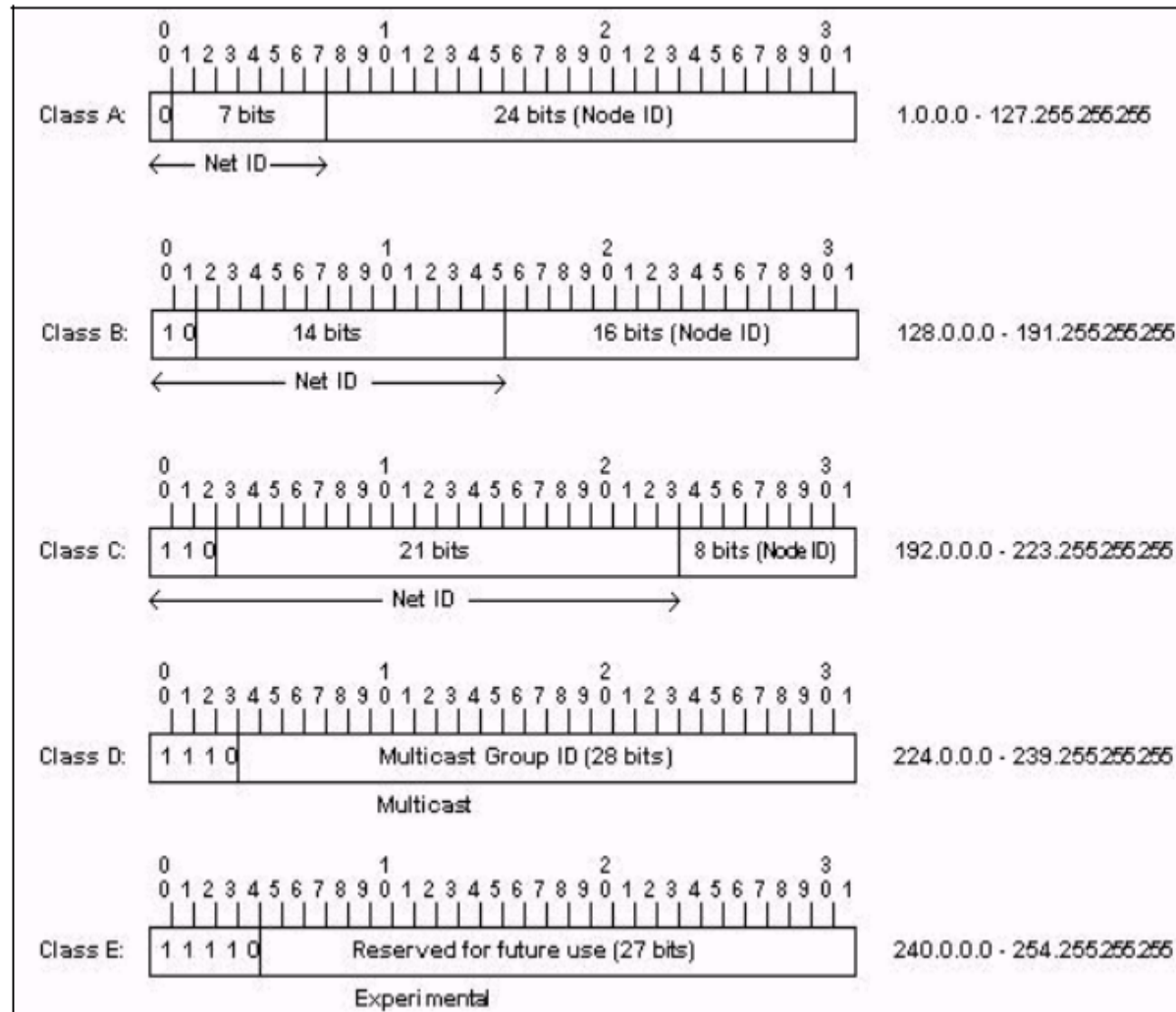
- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Example

Solution

- a. The first bit is 0. This is a class A address.***
- b. The first 2 bits are 1; the third bit is 0. This is a class C address.***
- c. The first byte is 14; the class is A.***
- d. The first byte is 252; the class is E.***

IP Address Classes



IP Address Classes

- Class A: The first octet is the network portion. Octets 2, 3, and 4 are for subnets/hosts
- Class B: The first two octets are the network portion. Octets 3 and 4 are for subnets/hosts
- Class C: The first three octets are the network portion. Octet 4 is for subnets/hosts

Private Address Range

Address Class	Reserved Address Space
Class A	10.0.0.0 - 10.255.255.255
Class B	172.16.0.0 - 172.31.255.255
Class C	192.168.0.0 - 192.168.255.255

Network Masks

- Distinguishes which portion of the address identifies the network and which portion of the address identifies the node.
- Default masks:
 - Class A: 255.0.0.0
 - Class B: 255.255.0.0
 - Class C: 255.255.255.0